

Lab 2 - Robot Operating System: Part B

2.1 Run your code with a new world map

In this lab, you will play with your robot in a new world map described in your Assignment.

Task 1 Build your node to drive the robot vertically in a new world in Gazebo.

- 1) Use the following commands to create *tutorial_turtlebot.launch.py*

```
cd ~/M-Drive/ros2_ws/src/tutorial_pkg/launch
gedit tutorial_turtlebot.launch.py
```

- 2) Copy the below code to the new *tutorial_turtlebot.launch.py* file

```
#!/usr/bin/env python3
import os
from launch import LaunchDescription
from launch.actions import DeclareLaunchArgument, IncludeLaunchDescription
from launch.conditions import IfCondition
from launch.launch_description_sources import PythonLaunchDescriptionSource
from launch.substitutions import LaunchConfiguration, EnvironmentVariable
from launch_ros.actions import Node
from launch_ros.substitutions import FindPackageShare

def generate_launch_description():

    # Constants for paths to different files and folders
    package_name = 'turtlebot3_gazebo'
    world_file_path = 'worlds/rosbot_sim_room_12.world'
    launch_file_path = 'launch'

    # Pose
    x_pose = '0.0'
    y_pose = '0.0'
    z_pose = '0.0'
    yaw_pose = '0.00'

    pkg_gazebo_ros = FindPackageShare(package='gazebo_ros').find('gazebo_ros')
    pkg_share = FindPackageShare(package=package_name).find(package_name)
    gazebo_world_path = os.path.join(pkg_share, world_file_path)
    launch_dir_path = os.path.join(pkg_share, launch_file_path)

    use_sim_time = LaunchConfiguration('use_sim_time', default='true')
    # Start Gazebo server
    start_gazebo_server_cmd = IncludeLaunchDescription(
        PythonLaunchDescriptionSource(os.path.join(pkg_gazebo_ros, 'launch', 'gzserver.launch.py')),
        launch_arguments={'world': gazebo_world_path}.items())

    # Start Gazebo client
    start_gazebo_client_cmd = IncludeLaunchDescription(
        PythonLaunchDescriptionSource(os.path.join(pkg_gazebo_ros, 'launch', 'gzclient.launch.py')),
    )
```

```

# Start the Robot State Publisher
start_robot_state_publisher_cmd = IncludeLaunchDescription(
    PythonLaunchDescriptionSource(
        os.path.join(launch_dir_path, 'robot_state_publisher.launch.py')
    ),
    launch_arguments={ 'use_sim_time': use_sim_time }.items()
)

# Launch the robot
spawn_turtlebot_cmd = IncludeLaunchDescription(
    PythonLaunchDescriptionSource(
        os.path.join(launch_dir_path, 'spawn_turtlebot3.launch.py')
    ),
    launch_arguments={
        'x_pose': x_pose,
        'y_pose': y_pose,
        'z_pose': z_pose,
        'yaw_pose': yaw_pose
    }.items()
)

return LaunchDescription([
    Node(
        package='tutorial_pkg',
        executable='tutorial_pkg_node',
        name='tutorial_pkg',
        output='screen'
    ),
    start_gazebo_server_cmd,
    start_gazebo_client_cmd,
    start_robot_state_publisher_cmd,
    spawn_turtlebot_cmd,
])

```

Then save the file *tutorial_turtlebot.launch.py* and exit.

- 3) Use the following commands to build it:

```

cd ~/M-Drive/ros2_ws
colcon build

```

- 4) Launch your node with the following commands:

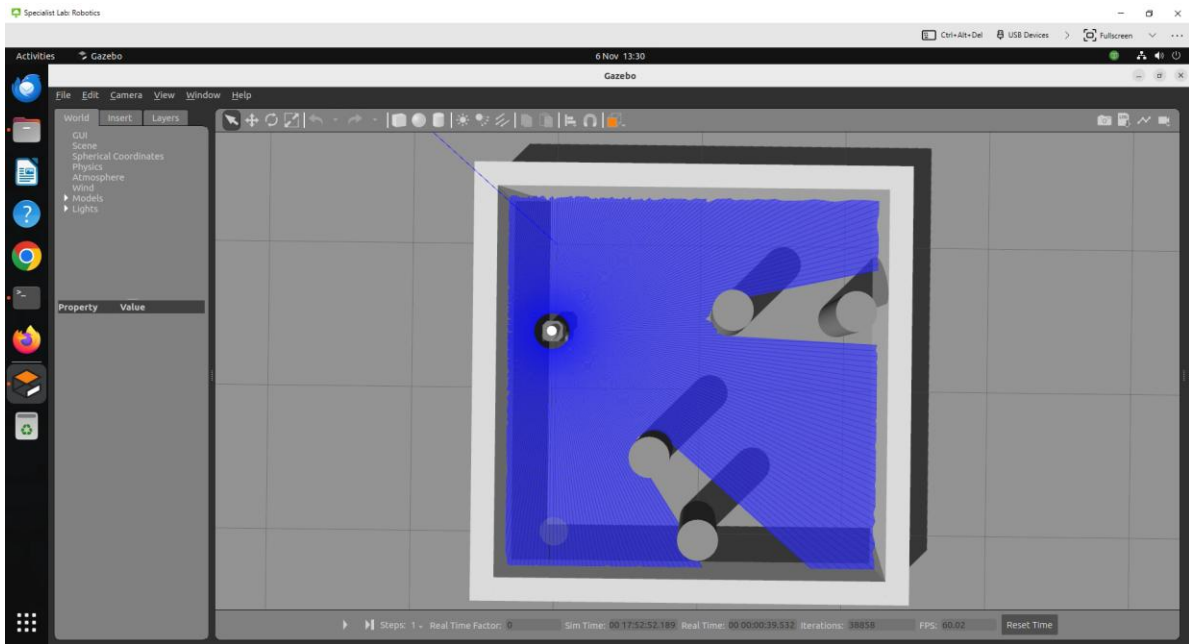
```

cd ~/M-Drive/ros2_ws
source install/setup.sh
export TURTLEBOT3_MODEL=burger
ros2 launch tutorial_pkg tutorial_turtlebot.launch.py

```

You will see the text movements displayed in the terminal.

After few seconds, a Gazebo simulator window will show Turtlebot moves vertically, as shown in the figure below.



5) Finally, you may key in Ctrl+C in the terminal to stop it.

2.2 Change the robot moving direction in Gazebo

Now, you may control the robot to move horizontally.

Task 2: Build your node to drive the robot horizontally in a new world in Gazebo.

1) Open [tutorial_turtlebot.launch.py](#) and change the following line:

```
# Pose
x_pose = '0.0'
y_pose = '0.0'
z_pose = '0.0'
yaw_pose = '1.57'
```

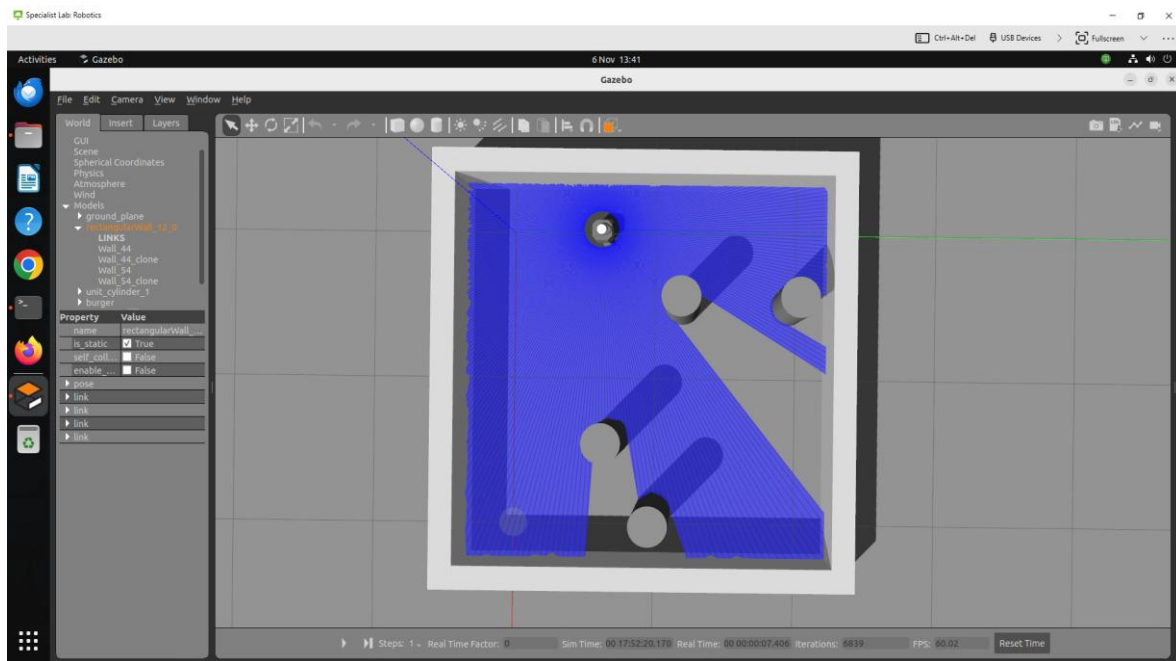
2) Use the following commands to build your workspace with the recent changes:

```
cd ~/M-Drive/ros2_ws
colcon build
```

3) Now, you can use the following commands to run your code.

```
cd ~/M-Drive/ros2_ws
source install/setup.sh
export TURTLEBOT3_MODEL=burger
ros2 launch tutorial_pkg tutorial_turtlebot.launch.py
```

4) After several seconds, a Gazebo simulator window will appear to show that Turtlebot moves horizontally, as shown in the figure below.



5) Finally, you may key in Ctrl^C in the terminal to stop it.